

Diego Martínez Calzada

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Junior Software Engineer and Computer Science graduate with strong analytical skills and eager to keep learning. Solid foundation in Python, cloud architectures, and AI applications. Passionate about designing scalable solutions, and implementing emerging technologies.

Education

Faculty of Science, UNAM

Bachelor of Science in Computer Science. GPA 3.9/4.0 (9.6/10)

Coyoacán, CDMX

February 2025

Certifications

AWS Certified Cloud Practitioner

AWS re/Start Program

February 2026 – Present

Technical Skills & Projects

Programming languages: Python, JavaScript

Frontend: React, Tailwind CSS, HTML, CSS

Backend: Django REST Framework

AI/ML: RAG systems, LlamaIndex, scikit-learn, Matplotlib, Pandas

Cloud: AWS, GCP

Tools: GitHub, GitHub Pages

AWS EC2 AI Technical Assistant | [<https://github.com/diegotlou/aws-rag-system>] | [[Interactive app](#)]

- Developed and deployed a web application in Python using Streamlit, integrating a hierarchical RAG system (LlamaIndex) with HyDE (Hypothetical Document Embeddings) to query cloud technical documentation.
- Integrated Pinecone as a vector database to enable high-speed semantic search capabilities and MongoDB for document storage achieving an easily scalable system.
- Evaluated response quality using faithfulness (5/5), relevance (4.9/5), and robustness (3.75/5) metrics, achieving an average rating of 91% on AWS EC2 documentation.

Personal portfolio | [<https://github.com/diegotlou/diegotlou-portfolio/>] | [[Portfolio](#)]

- Built a responsive developer portfolio with React and Tailwind CSS, featuring reusable components and Motion animations, deployed via GitHub Pages, ensuring an intuitive and fluid experience.
- Achieved Lighthouse scores of 97 Performance, 100 Best Practices, 91 SEO and 82 Accessibility.

Experience

ICAT (Institute of Applied Science and Technology)

Remote

Software developer (social service)

August 2023 – March 2024

- Optimized UI components to improve load performance and long-term maintainability of the Muro Colaborativo platform.
- Debugged and resolved AJAX request failures, restoring expected data flow between frontend and backend services.
- Collaborated with the project team to define the platform requirements and write the complete technical documentation to significantly reduce the onboarding time for new developers.